

Equal-size generating sets give different Cayley-core edge counts

Disproof of Conjecture 00000003467

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Abstract

On the same cyclic group $\mathbb{Z}/12\mathbb{Z}$, the inverse-closed generating sets $S_1 = \{\pm 1, \pm 3\}$ and $S_2 = \{\pm 1, \pm 2\}$ both have size 4. Their connected Cayley graphs have cores K_2 and K_3 , respectively, with 1 and 3 edges. Consequently the core edge count is not a function of generating-set size even when the group is fixed. This disproves the asserted linear dependence in Conjecture 00000003467, independently of its specified slope.

Definitions and scope

We use finite simple undirected graphs. For an additive group G and a subset $S = -S$ not containing 0, the Cayley graph has vertex set G and adjacency $x \sim y$ if $y - x \in S$. A graph homomorphism preserves adjacency. A core of a finite graph is a vertex-minimal subgraph to which the graph has a homomorphism. Equivalently, it is a retract whose endomorphisms are injective. The standard Cayley and core conventions are described in Roberson, Introduction and Section 1. We give the necessary correspondence below.

The source asserts a linear dependence on generating-set size for abelian groups. We keep the entire group fixed, so its rank and every other group invariant remain fixed. Both connection sets below genuinely generate the group and the graphs are connected. The source does not require generating sets to be irredundant.

Two Cayley graphs on the same group

Let $G = \mathbb{Z}/12\mathbb{Z}$ and put

$$S_1 = \{1, 11, 3, 9\} = \{\pm 1, \pm 3\}, \quad S_2 = \{1, 11, 2, 10\} = \{\pm 1, \pm 2\}.$$

Both sets have four elements, omit 0, and are closed under negation. Each contains 1, which generates G . In either graph, repeatedly following the edge $x \sim x + 1$ reaches every vertex, proving connectedness. Write $A = \text{Cay}(G, S_1)$ and $B = \text{Cay}(G, S_2)$.

For A , take the induced edge on vertices $\{0, 1\}$ and define

$$r_2(x) = x \bmod 2 \in \{0, 1\}.$$

This is well-defined on $\mathbb{Z}/12\mathbb{Z}$ because 2 divides 12. Every difference in S_1 is odd, so adjacent vertices have distinct images. Thus r_2 is a graph homomorphism to this copy of K_2 . It fixes 0 and 1, making it a retraction.

For B , the vertices $\{0, 1, 2\}$ form an induced triangle: all their nonzero differences lie in S_2 . Define

$$r_3(x) = x \bmod 3 \in \{0, 1, 2\}.$$

This is well-defined since 3 divides 12. None of 1, 11, 2, 10 is zero modulo 3, so r_3 preserves adjacency to K_3 . It fixes the three triangle vertices and is therefore a retraction.

Why these retracts are the actual cores

Lemma 1. *If a finite graph X retracts onto a complete graph K_t , then K_t is its core, unique up to graph isomorphism.*

Proof. Every homomorphism out of K_t is injective: identifying two vertices would turn their edge into a loop. If X maps to a subgraph Y , the restriction to the displayed K_t therefore forces $|V(Y)| \geq t$. Since the retraction onto K_t attains this bound, it is a core. For a vertex-minimal choice Y , the same restriction is an injection from K_t onto its t vertices. It preserves every edge of K_t , so Y is complete and isomorphic to K_t . $\square$

Applying the lemma to the explicit retractions yields

$$\text{core}(A) \cong K_2, \quad \text{core}(B) \cong K_3.$$

In particular their edge counts are exactly 1 and 3; they are not merely upper bounds inferred from colorings.

Theorem 1. *The abelian core-edge-count assertion of Conjecture 00000003467 is false.*

Proof. If a function F_G of generating-set size determined the core edge count for this fixed group, the two examples would give

$$F_G(4) = 1, \quad F_G(4) = 3,$$

a contradiction. Thus no such function exists, in particular no linear or affine formula with slope half the group rank. Failure of the abelian assertion already disproves the conjunction; no statement about the separate nonabelian bound is needed. $\square$

Formal correspondence

The Lean project constructs the actual ring `ZMod 12` and two `SimpleGraph` objects directly from the difference-membership Cayley definition. It checks the generating-set sizes, proves their additive subgroup closures are the whole group, and proves graph connectedness by walks of unit steps. The quotient-color maps and inclusions are actual graph homomorphisms, and their left-inverse identities are checked.

The local predicate `IsCore` is the standard injective-endomorphism condition; `IsCoreOf` adds an explicit retraction. The theorem `retraction_reflects_adjacency` checks that the inclusion identifies an induced subgraph. `core_minimal` proves the minimum-vertex property against any finite subgraph candidate, so the core predicate is connected to the usual definition rather than merely naming a convenient retract.

The formal uniqueness proof also covers every finite core satisfying this standard predicate. If H is another core and there are homomorphisms $f : H \rightarrow K_t$ and $g : K_t \rightarrow H$, the endomorphism gf is injective, hence surjective by finiteness. Consequently g is surjective; it is also injective because its source is complete. It therefore identifies H with K_t as a graph. This is proved in `complete_core_unique`, and `complete_core_edge_count` transfers the actual finite edge count to any such core. The final counterexample and `no_function_of_generator_count` use these verified cores and Mathlib's unordered edge sets.

Reproduction

Lean 4.19.0 and Mathlib revision

`c44e0c8ee63ca166450922a373c7409c5d26b00b` are pinned. In `lean/`, run

```
lake exe cache get
lake build
lake env lean -DwarningAsError=true Main.lean
```

The cache command fetches pinned official dependencies if absent. The submitted project itself is built in a fresh directory, with warnings treated as errors. Finite calculations are kernel checked; no external numerical oracle, extra axiom or proof placeholder is used. No auxiliary numerical program is required. Run `tectonic main.tex` to regenerate the PDF. The author performed a separate adversarial self-review; no independent reviewer or subagent was used. Exact source, provenance, build logs, axiom dependencies, hashes and review records accompany this submission.

References

1. The Last Math Competition, Conjecture 00000003467. The exact bilingual source is reproduced in `SOURCE.md`.
2. D. E. Roberson, *Cores of Vertex Transitive Graphs*, Electronic Journal of Combinatorics 20(2) (2013), P45, Introduction and Section 1. Publisher's full text.